Every Denominator Occurs in the Mean of the Divisors with Exact Minima at Prime-Power Indices

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Abstract

For every positive integer n there is an integer K for which the reduced denominator of $\sigma(K)/\tau(K)$ is exactly n. This proves the existence conjecture recorded in OEIS A346644. For every odd prime q and every $s \geq 1$, the least such K at denominator q^s is given by a finite minimum over the partitions of s, using the first primes not congruent to 1 modulo q. The remaining prime-power family has the closed formula $a(2^s) = 2^{2^s-1}$. In particular $a(q) = 2^{q-1}$ for every prime q.

1 A witness for every denominator

Let $\sigma(K)$ be the sum of the positive divisors of K and $\tau(K)$ their number. Define

$$a(n) = \min\{K \geq 1 : \operatorname{den}(\sigma(K)/\tau(K)) = n\}.$$

OEIS A346644 [1] records this sequence with zero as an escape value and conjectures that every index has a witness.

Theorem 1 (Universal denominator construction)**.** *Write $n = 2^a m$ with m odd, and put*

$$u = 2^a, \qquad A = 2^u - 1, \qquad h = mA.$$

For any odd prime $p \equiv -1 \pmod{n}$, the integer

$$K = 2^{u-1} p^{h-1} \tag{1}$$

satisfies $\operatorname{den}(\sigma(K)/\tau(K)) = n$. Such a prime exists for every n.

Proof. Dirichlet's theorem [2] supplies infinitely many primes $p \equiv -1 \pmod{n}$, since $\gcd(-1, n) = 1$; choose an odd one. For $n = 1$ the congruence is vacuous. The factors in (1) are coprime, so

$$\tau(K) = uh = umA = nA, \qquad \sigma(K) = AS, \quad S = 1 + p + \cdots + p^{h-1}.$$

The integer h is odd. Reducing the alternating sum modulo n gives $S \equiv 1 \pmod{n}$. Hence

$$\frac{\sigma(K)}{\tau(K)} = \frac{S}{n}, \qquad \gcd(S, n) = 1.$$

This also covers $n = 1$, for which $u = A = h = K = 1$. □

2 Exact least values at every prime power

For a prime q, write v_q for its valuation. A factor p^e of K contributes

$$c_q(p, e) = v_q(e + 1) - v_q(1 + p + \cdots + p^e)$$

to the denominator valuation before taking its positive part.

Lemma 1. *If $c_q(p,e) > 0$, then $p \not\equiv 1 \pmod q$ and $c_q(p,e) \le v_q(e+1)$. For any prime $p \not\equiv 1 \pmod q$ and integer $b \ge 1$,*

$$\operatorname{den}\left(\frac{1+p+\cdots+p^{q^b-1}}{q^b}\right) = q^b. \tag{2}$$

Proof. The inequality follows immediately from the nonnegative numerator valuation. If q is odd and $p \equiv 1 \pmod q$, lifting the exponent gives $v_q((p^{e+1}-1)/(p-1)) = v_q(e+1)$, so $c_q(p,e) = 0$. For $q = 2$ and odd p, if e is even both valuations are zero; if e is odd, the elementary 2-adic lifting formula gives

$$v_2((p^{e+1}-1)/(p-1)) = v_2(p+1) + v_2(e+1) - 1 \ge v_2(e+1).$$

Thus positive contribution again requires $p \not\equiv 1 \pmod q$. Finally $p^{q^b} \equiv p \pmod q$ by Fermat's theorem, and $p-1$ is invertible modulo q. The numerator sum in (2) is therefore 1 modulo q, proving the asserted reduced denominator. □

Theorem 2 (Partition formula)**.** *Fix an odd prime q, and list the primes not congruent to 1 modulo q as $\ell_1(q) < \ell_2(q) < \cdots$. For every $s \ge 1$,*

$$a(q^s) = \min_{\substack{b_1 \ge \cdots \ge b_t \ge 1 \\ b_1+\cdots+b_t = s}} \prod_{i=1}^{t} \ell_i(q)^{q^{b_i}-1}. \tag{3}$$

The minimum is over the ordinary integer partitions of s. For $q = 2$,

$$a(2^s) = 2^{2^s-1} \qquad (s \ge 0). \tag{4}$$

Consequently $a(q) = 2^{q-1}$ for every prime q.

Proof. Every candidate in (3) has divisor count q^s and divisor sum coprime to q, by (2). Its reduced denominator is exactly q^s.

Conversely, let $K = \prod p_i^{e_i}$ have reduced denominator q^s. The sum of all $c_q(p_i, e_i)$ is s, so the sum over its positive terms is at least s. By Lemma 1, the corresponding primes are admissible and their values $a_i = v_q(e_i+1)$ have sum at least s. Choose integers $0 \le b_i \le a_i$ with total s, discard the zeros, and replace each exponent by $q^{b_i}-1 \le e_i$. The resulting integer $K' \le K$ has denominator exactly q^s by (2). Thus it suffices to minimize such products.

For a fixed list of positive b_i, use the smallest admissible primes and assign larger exponents to smaller primes. The latter follows from $x^A y^B \le x^B y^A$ when $x \le y$ and $A \ge B$. This gives (3). There are sufficiently many admissible primes by Dirichlet's theorem, applied to -1 modulo the odd prime q.

For $q = 2$, the only admissible prime is 2. The same replacement argument requires the exponent of 2 to be at least $2^s - 1$. The integer 2^{2^s-1} attains denominator 2^s, since its divisor sum is odd; this proves (4), including $a(1) = 1$. For odd prime q, the only partition of 1 is (1) and $\ell_1(q) = 2$. □

For example $a(9) = \min(2^8, 2^2 3^2) = 36$ and $a(25) = \min(2^{24}, 2^4 3^4) = 1296$. At denominator 81 the partition $(2,1,1)$ gives $2^8 3^2 5^2 = 57600$, smaller than the value 108900 from $(1,1,1,1)$; comparison with the other three partitions proves $a(81) = 57600$.

The supplied verifier independently computes divisor sums and counts from integer factorizations, enumerates initial minima, checks the published values, and evaluates (3). Large constructive witnesses are checked modulo their symbolic divisor count, avoiding expansion of enormous powers.

References

[1] B. Cloitre, OEIS A346644, https://oeis.org/A346644, accessed October 26, 2026.

[2] P. G. L. Dirichlet, *There are infinitely many prime numbers in all arithmetic progressions with first term and difference coprime* (1837), translated by R. Stephan, arXiv:0808.1408v2, https://arxiv.org/abs/0808.1408v2.